Evaluating Machine Learning Performance and Limitations for Imbalanced NHL Shot Event Classification

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Abstract

This study is based on the shooting data of the NHL for the 2020-2025 season. We compared and analyzed the performance of logistic regression and various models in predicting goal events. We found that HGB and XGB achieved the best results in terms of overall accuracy and ROC-AUC. This project also indicates the severe challenges faced by the current NHL goal event prediction task. Due to the imbalance of categories and the simplicity of feature dimensions, the models still have deficiencies in distinguishing shooting events from actual goals. The GitHub repository for the project: https://github.com/honzh073/Sports

1 Introduction

In NHL games, most previous studies have focused on expected goals. This refers to the number of goals a team is expected to score under specific conditions (Macdonald, 2012), such as goals scored during regulation time. The expected goals statistic serves multiple purposes. For example, it can be used to evaluate a team's offensive and defensive performance throughout a full season, as well as to assess individual player performance. The study (Macdonald, 2012) constructed an expected goals model using methods such as ridge regression, incorporating metrics including shots on goal, missed shots, blocked shots, hits, and face-offs. The study (Wang and Zhang, 2012) modeled and classified ice hockey shooting events (video) using a mixed hidden Markov model. The study (Michaud-Paquette et al., 2009) employed multiple regression analysis combined with kinematic features—such as shot direction and velocity, shaft flex, and blade orientation—to predict shooting accuracy. However, this study adopted a stationary shooting setup and did not account for variations in shooting angle or distance to the goal.

During a game, shooting events occur very rapidly. The dynamic angle, distance, and technique used in a shot may play a role in predicting or classifying goal events. Our project aims to leverage spatial coordinate information and shooting techniques to predict goal events (probabilistically) or classify them (by type).

2 Background

The National Hockey League (NHL) is a professional ice hockey league in North America consisting of 32 teams, 25 of which are based in the United States and 7 in Canada (Wikipedia, 2026c). The NHL is the world's top professional ice hockey league and is also the fifth-highest-grossing professional sports league globally (Wikipedia, 2026c).

During the game, a large amount of event-based play-by-play data and event records were generated. The NHL data records often cover several decades. For instance, the MoneyPuck website (MoneyPuck, 2026) has recorded all NHL game data from the 2007 season to the 2024 season. These data provide important indicators for the analysis of ice hockey.

2.1 NHL Rink Surface

According to the Official Rules 2024-2025 document of the NHL (National Hockey League, 2024), we can learn that the size of the NHL ice rink is 200 feet by 85 feet.

From the Figure 1, we can see that the ice surface is demarcated by two blue lines, which divided the rink into three zones: the defensive zone (D), extending from the goal line to the team's own blue line; the neutral zone (N), between the two blue lines; and the offensive zone (O), crossing from the opposing team's blue line to the opposing goal line.

Following the standard NHL rink dimensions on official rule book (National Hockey League, 2024), layout Figure 1 set center ice as the origin, the goal positions are as follows: the home team's goal is located at (-89, 0) ft (defensive end), and the away team's goal is located at (89, 0) ft (offensive end).

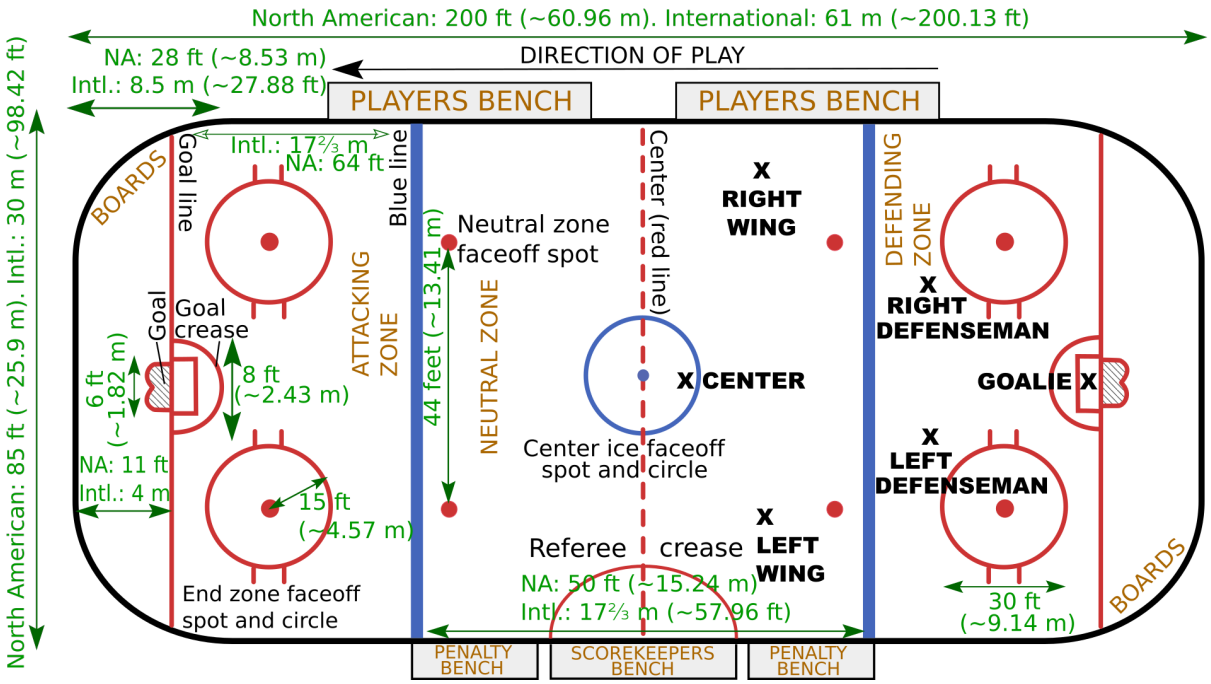

Figure 1: The spatial distribution and dimensions of the ice hockey rinks in North America/NHL. Image source: https://commons.wikimedia.org/wiki/File:Ice_hockey_layout.svg.

2.2 Players allocation and Goals rule

An NHL team fields six players on the ice: one goaltender, two defensemen (left and right), and three forwards (center, left wing, and right wing). A goal is scored or recorded when the puck completely crosses the goal line between the goal posts and beneath the crossbar (Wikipedia, 2026a).

The above are the general rules of the NHL. The official rules of the NHL (National Hockey League, 2024) contain more detailed regulations. In summary, the playing fields in the NHL are more compact and full of intense competition. Understanding these background facts will help one better comprehend this project.

3 Data

3.1 Data Description

The NHL clean shots data we use is sourced from Kaggle (Samuarg, 2026). The file type of this dataset is CSV, the name of the dataset is "nhl_shots_data.csv". This dataset contains 528,720 rows of shooting records and 19 feature columns. This dataset covers five regular seasons from the 2020-2021 season to the 2024-2025 season. The dataset contains cleaned and recorded individual NHL team shot data. This dataset includes spatial information, including the precise X and Y coordinates of each shot. This dataset provides detailed event information. Unlike the conventional score sheet, this dataset contains:

- **Spatial coordinates**: X and Y coordinates relative to the center of the ice surface;
- **Event information**: Specific area like offensive zone, neutral zone, defensive zone and the side of the home team's defense;
- **Shot Record**: Different types of shots like wrist, slap, tip-in, snap and wrap-around.

3.2 Data Format

Each collected shot record is structured and stored as a CSV object. An example of the data schema is illustrated below:


```
{
  "gameId": 2024020001,
  "gameDate": 2024-10-04,
  "season": 20242025,
  "homeTeam": BUF,
  "awayTeam": awayTeam,
  "homeTeamId": 7,
  "awayTeamId": 1,
  "eventType": shot-on-goal,
  "period": 1,
  "periodTimeRemaining": 19:52,
  "homeTeamDefendingSide": right,
  "zoneCode": O,
```

```
  "xCoord": 56,
  "yCoord": -39,
  "shotType": wrist,
  "shootingPlayerId": 8483495,
  "goalieInNetId": 8480045.0,
  "eventOwnerTeamId": 1,
  "sortOrder": 13
}
```


The following is the description of these 19 columns, including their specific value ranges and the proportion part of the types (Samuarg, 2026):

- **gameId**: The unique numerical identifier for the specific game.
- **gameDate**: The date the game was played (YYYY-MM-DD).
- **season**: The identifier for the specific NHL season.
- **homeTeam**: The abbreviation for the home team.
- **awayTeam**: The abbreviation for the away team.
- **homeTeamId**: The unique identifier for the home team.
- **eventType**: The class of event (63% shot-on-goal, 29% missed-shot, 7% Goal).
- **period**: The period of play (1, 2, 3, 4 for OT, or 5 for Shootouts).
- **periodTimeRemaining**: The time remaining in the current period (MM:SS).
- **homeTeamDefendingSide**: The side of the rink the home team is defending (51% right or 49%left).
- **zoneCode**: The zone where the event occurred (96% O = Offensive, 3% N = Neutral, 1% D = Defensive).
- **xCoord**: The X-coordinate of the puck in feet (-100 to +100).
- **yCoord**: The Y-coordinate of the puck in feet (-42 to +42).
- **shotType**: The types of the shot (53% wrist, 15% snap, 31% Other).

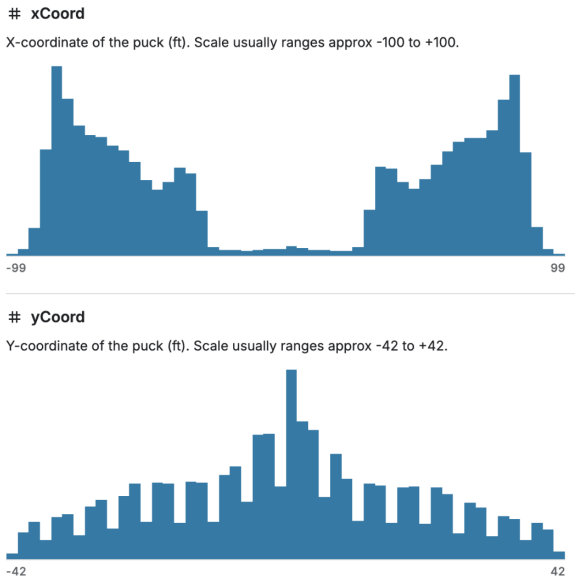

Figure 2: X coordinate of the puck (ft) ranges approx -100 to +100. Y coordinate of the puck (ft) ranges approx -42 to +42 (Samuarg, 2026).

- **shootingPlayerId**: The unique identifier for the player taking the shot.
- **goalieInNetId**: The unique identifier for the goalie defending the net.
- **eventOwnerTeamId**: The identifier for the team that initiated the event.
- **sortOrder**: The chronological sequence number of the event within the game.

The dataset's Kaggle page provides detailed explanations for these columns along with the distribution of their values. We take xCoord and yCoord as an example.

3.3 Data Preprocessing

We examined each column in the dataset to identify any special values or missing values that might be present within it. To facilitate data processing, maintain feature consistency, and handle missing values in the columns, we carried out two main preprocessing steps on the dataset.

- **Time feature transformation**: The periodTimeRemaining (MM:SS) column in the original string format is converted into a numerical representation in total seconds. We apply the transformation: total seconds = (M × 60) + S.
- **Random Imputation for missing shots types**: The shotType column contains 66 missing entries. To maintain the integrity of the

dataset, we perform random imputation for these missing values. Each missing record is replaced with a value sampled from the known set of shooting types: wrist, slap, tip-in, snap, wrap-around, backhand, poke, deflected, bat, cradle, between-legs.

3.4 Unbalanced Data

The number of instances in each class is not equal, and this distribution presents challenges for classification on the dataset. Minority class instances are typically represented as positive examples, while majority class instances are represented as negative examples. This is a very common challenge when minority class instances are insufficient (Wang et al., 2021), as is the case with the "goal" event we focus on in the NHL dataset. The imbalance in the NHL dataset in this project is primarily characterized by class imbalance, resulting from the small number of instances labeled as "goal" (approximately 7%).

Figure 3 illustrates the distribution of event types in the NHL dataset that "goal" events are underrepresented compared to "shot-on-goal" and "missed-shot" events.

As shown in Figure 4, the distribution of event types across different team sides (home vs. away) and zones (offensive, neutral, defensive) in the NHL dataset. the left panel shows that both home and away teams have a similar distribution of event types, while the right figure shows that the majority of events occur in the offensive zone.

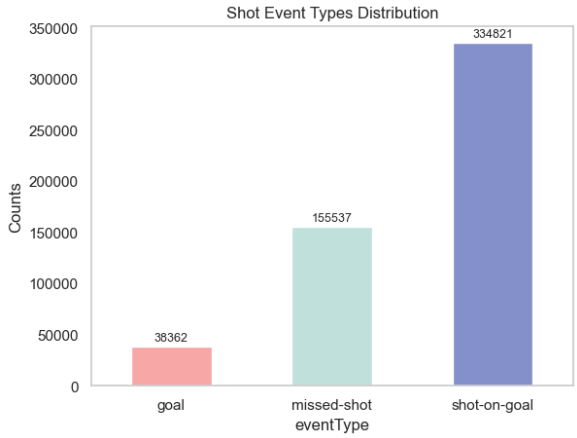

Figure 3: Class distribution of events in the NHL dataset, showing the underrepresentation of the "goal" class.

3.5 Feature Engineering and Encoding

Feature engineering is a process of feature construction and transformation in machine learning (Kilaru and Cheemakurthi, 2021). It preprocesses the raw data to make it more suitable for the requirements of the prediction model and enhances the interpretability of the model (Amershi et al., 2019).

We convert the original physical coordinates into spatial geometric features. By combining the home or away status of the team and the defensive direction during the half (homeTeamDefendingSide), we determine the position of the center point of the target goal and calculate the shooting distance (shotDistance) and shooting angle (shotAngle). Among them, the shooting distance is defined using the Euclidean distance formula $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. Through this transformation, we map the coordinate points to describe physical indicators such as shooting distance and angle.

We also transformed the qualitative descriptive features such as eventType, zoneCode, and shotType into discrete integer indices through a predefined mapping dictionary. While preserving the original classification logic of the dataset, we converted the non-numeric game records into structured features that can be directly processed by the machine learning model.

4 Algorithms

In this section, we will introduce the algorithms we have used: Logistic Regression, Random Forest Classifier, Gradient Boosting Classifier and XGBoost.

4.1 Logistic Regression

Logistic Regression is a standard probabilistic statistical classification model (Feng et al., 2014). The prediction result of Logistic Regression for a sample is the probability that the sample belongs to the positive class or the negative class (Feng et al., 2014).

Logistic regression maps a linear combination of features into a probability range between 0 and 1 using the sigmoid function (Bisong, 2019):

$$h(t) = \frac{1}{1 + e^{-t}}. \tag{1}$$

The logistic regression model is driven by the sigmoid function, which is formally represented as follows:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n \tag{2}$$

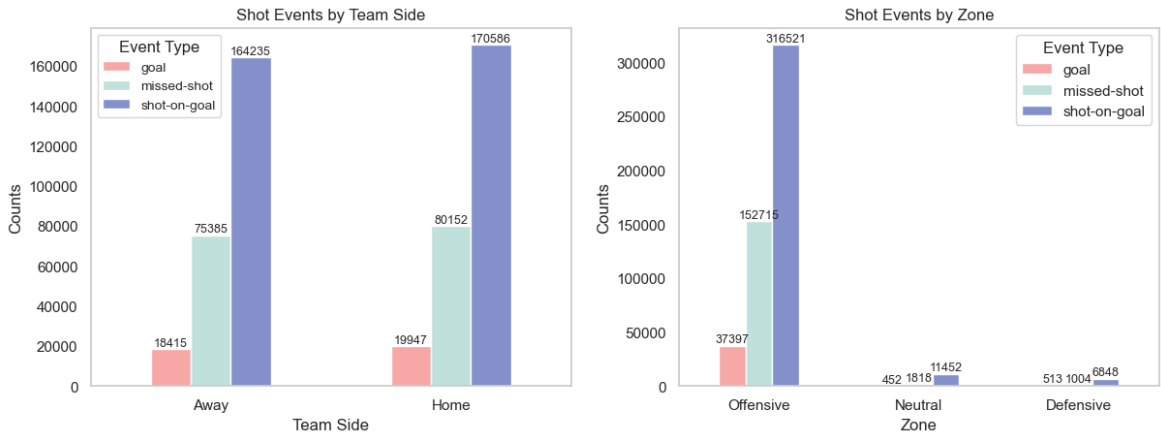

Figure 4: Distribution of event types categorized by team side (left) and rink zone (right).

$$h(\hat{y}) = \frac{1}{1 + e^{-\hat{y}}} \tag{3}$$

To construct a classification model with K classes, the multinomial logistic model takes the following formal definition:

$$\hat{y}(k) = \theta_0^k + \theta_1^k x_1 + \theta_2^k x_2 + \cdots + \theta_n^k x_n \tag{4}$$

$$p(k) = \frac{e^{\hat{y}(k)}}{\sum_{j=1}^{K} e^{\hat{y}(j)}} = \frac{e^{\theta_0^k + \theta_1^k x_1 + \cdots + \theta_n^k x_n}}{\sum_{j=1}^{K} e^{\theta_0^j + \theta_1^j x_1 + \cdots + \theta_n^j x_n}} \tag{5}$$

We chose logistic regression as the baseline model because it is simple and the probabilistic framework it employs is highly interpretable.

4.2 Random Forest Classifier

Random forest is composed of many decision tree classifiers that are completely independent of each other. When a test sample is input, the final category label of the sample is determined by the voting results of all the decision trees (Breiman, 2001). Figure 5 shows the classification process based on random forest (Parmar et al., 2018).

Random forest achieves this by fitting multiple decision tree classifiers on different subsets of the dataset and using the averaging method to enhance prediction accuracy, while effectively controlling overfitting (scikit-learn, 2026d).

4.3 Gradient Boosting Classifier

Gradient boosting (Friedman, 2001) combines multiple weak "learners" into a strong learner through an iterative process (Wikipedia, 2026b). The goal

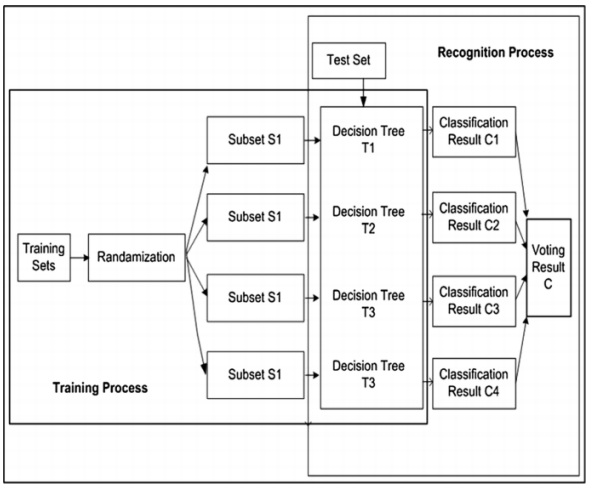

Figure 5: The framework of the random forest classifier (Breiman, 2001).

of gradient boosting is to minimize the loss function by iterating in the opposite direction of the gradient (Bahad and Saxena, 2019).

Starting with a model composed of constant functions, $F_0(\mathbf{x})$ gradually expands in a greedy manner (Wikipedia, 2026b; Friedman, 2001):

$$F_0(\mathbf{x}) = \arg\min_{\rho} \sum_{i=1}^{N} L(y_i, \rho) \tag{6}$$

- $F_0(\mathbf{x})$: The initial model.
- $L(y_i, \rho)$: The loss function measuring prediction error.
- ρ: The constant parameter to be optimized.
- N: The total number of training samples.

Then we could update the model by adding a new weak learner $h_m(\mathbf{x})$, scaled by the step size

γ_m, to the model from the previous iteration to minimize the remaining error:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \gamma_m h_m(\mathbf{x}) \tag{7}$$

We used the Histogram-based Gradient Boosting Classification Tree (scikit-learn, 2026b) and XGBoost (Chen and Guestrin, 2016) in our experiment, which is considerably faster when handling medium to large datasets.

- **HistGradientBoostingClassifier:** By applying the histogram-based technique, the efficiency and scalability of gradient boosting are enhanced, while the computational complexity and memory usage are reduced (GeeksforGeeks, 2026).

- **XGBoost:** XGBoost is an optimized distributed gradient boosting method. Within the framework of gradient boosting, XGBoost offers a parallel tree boosting algorithm (XGboost Documentation, 2026).

5 Test

We divided the preprocessed NHL shooting dataset into a training set and a test set. The dataset was randomly divided into an 80% training set and a 20% test set.

The features we use are: 'period', 'homeTeamDefendingSide', 'zoneCode', 'xCoord', 'yCoord'. 'shotType', 'periodTimeRemainingSec', 'isHomeTeam', 'shotDistance', 'shotAngle'.

As shown in the table 1, the training set contains 475,848 records, and the test set contains 52,872 records. We found that there is a significant class imbalance in the shooting events: approximately 63% are "Shot-on-goal", 29% are "Missed-shot", and the "Goal" category accounts for only about 7.2%. The distribution of event types in the training set and the test set is consistent.

To evaluate the performance of the model on the imbalanced dataset of NHL, we employed evaluation metrics such as balanced accuracy score, weighted F1 score, weighted recall rate, and weighted ROC-AUC (scikit-learn, 2026c,a).

6 Results

Table 2 presents the performance comparison of the four models: Logistic Regression (LR), Random Forest (RF), Histogram-based Gradient Boosting (HGB), and XGBoost (XGB). The evaluation metrics include precision, recall, F1-score for each class, as well as overall accuracy and ROC-AUC.

Based on the summary of the results in Table 2, we found that classifiers such as RF, HGB, and XGB outperformed the baseline logistic regression classifier in all overall evaluation metrics, demonstrating better classification performance. HGB and XGB performed better in terms of accuracy and ROC-AUC. HGB achieved the highest accuracy, while XGB achieved the best ROC-AUC values (0.6664). They demonstrated better performance in the NHL clean shots data (2020-2021 to 2024-2025) dataset.

7 Conclusion

The prediction or classification of NHL goal events is fraught with challenges, and the existing datasets are lacking in certain aspects. The classification results shows that HGB and XGB models outperform the logistic regression baseline in overall performance, but the overall prediction accuracy is not very good. The type of shooting events in NHL is influenced by multiple factors. In this dataset, only the spatial information of the ball is available, but factors such as the position of the players and other aspects have not been well recorded. Although the HGB and XGB models have improved the recall rate, the precision rate of the "goal" category is extremely low (0.15), indicating that the models have obvious limitations when dealing with imbalanced data and are difficult to distinguish between the types of shooting and actual scoring.

8 Future Work

In the future, improvement can be achieved through addressing the issue of data imbalance and conducting more in-depth feature extraction. In the future, the focus of predicting the type of goal events can be placed on addressing the issue of class imbalance, such as by introducing SMOTE sampling (imbalanced-learn developers, 2026). Carry out more in-depth feature extraction or utilize larger and more comprehensive data sets, for instance, conducting feature extraction based on information such as the distribution of players on the field, or using more comprehensive NHL data (MoneyPuck, 2026).

Dataset	Shot-on-goal (0)	Missed-shot (1)	Goal (2)	Total
Train (Count)	267,712	124,606	30,658	422,976
Train (%)	63.29%	29.46%	7.25%	100.00%
Test (Count)	67,109	30,931	7,704	105,744
Test (%)	63.46%	29.25%	7.29%	100.00%

Table 1: The data distribution of event types (80% training set, 20% test set)

Metric	Class / Avg	LR	RF	HGB	XGB
Precision	Shot-on-goal (0)	0.69	0.71	**0.72**	**0.72**
	Missed-shot (1)	0.34	**0.37**	**0.37**	**0.37**
	Goal (2)	0.13	**0.15**	**0.15**	**0.15**
Recall	Shot-on-goal (0)	**0.33**	0.31	0.32	0.32
	Missed-shot (1)	0.41	**0.48**	**0.48**	0.47
	Goal (2)	0.64	**0.70**	0.69	**0.70**
F1-score	Shot-on-goal (0)	0.44	0.43	**0.45**	0.44
	Missed-shot (1)	0.37	**0.42**	**0.42**	**0.42**
	Goal (2)	0.22	**0.24**	**0.24**	**0.24**
Summary	Accuracy	0.37	0.39	**0.40**	0.39
	ROC-AUC	0.6266	0.6637	0.6663	**0.6664**

Table 2: Classification reports comparison of Logistic Regression classifier (LR), Random Forest classifier (RF), Histogram-based Gradient Boosting classification tree (HGB), and XGB classifier (XGB).

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